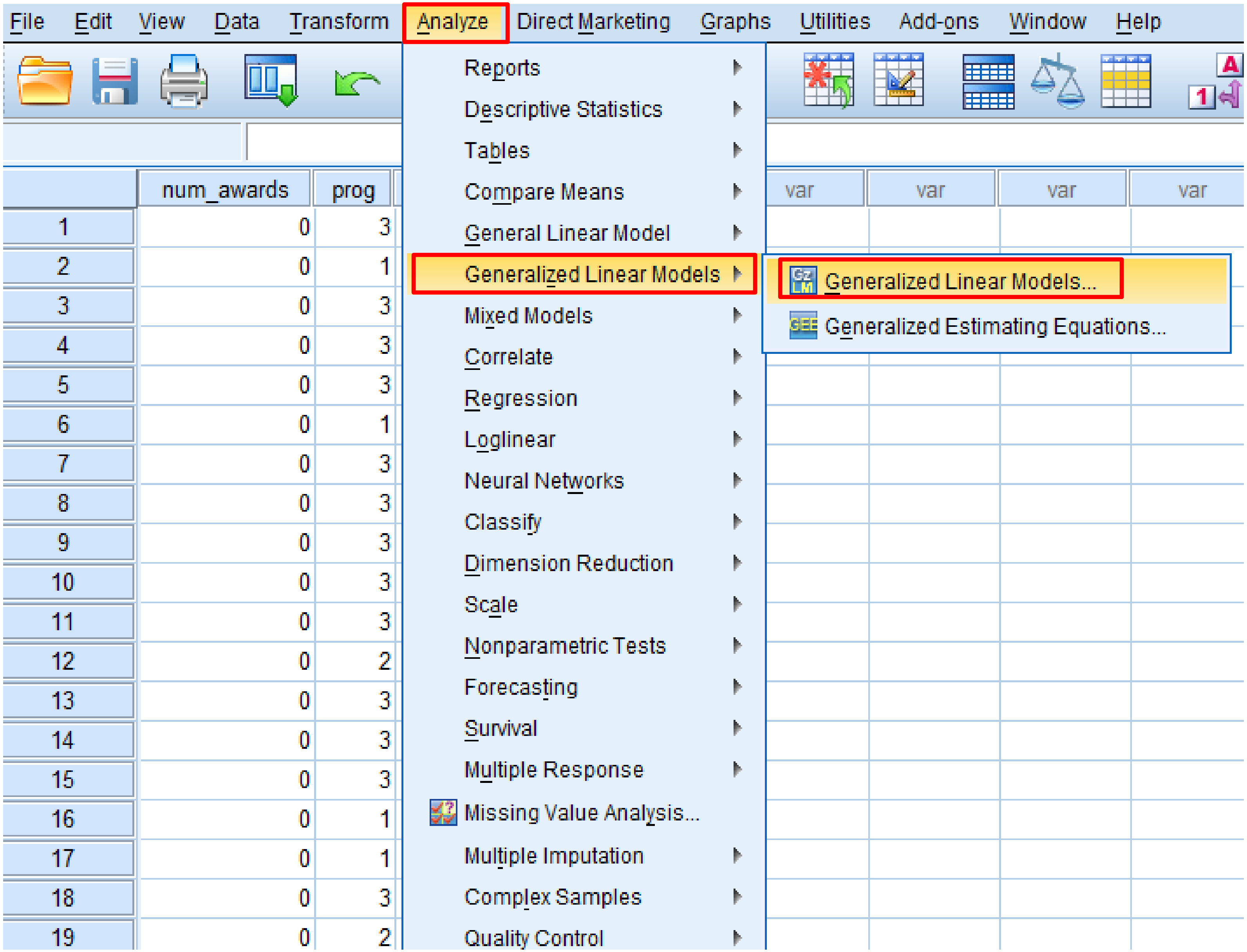
Poisson Regression – summary and tips for interpretation of output

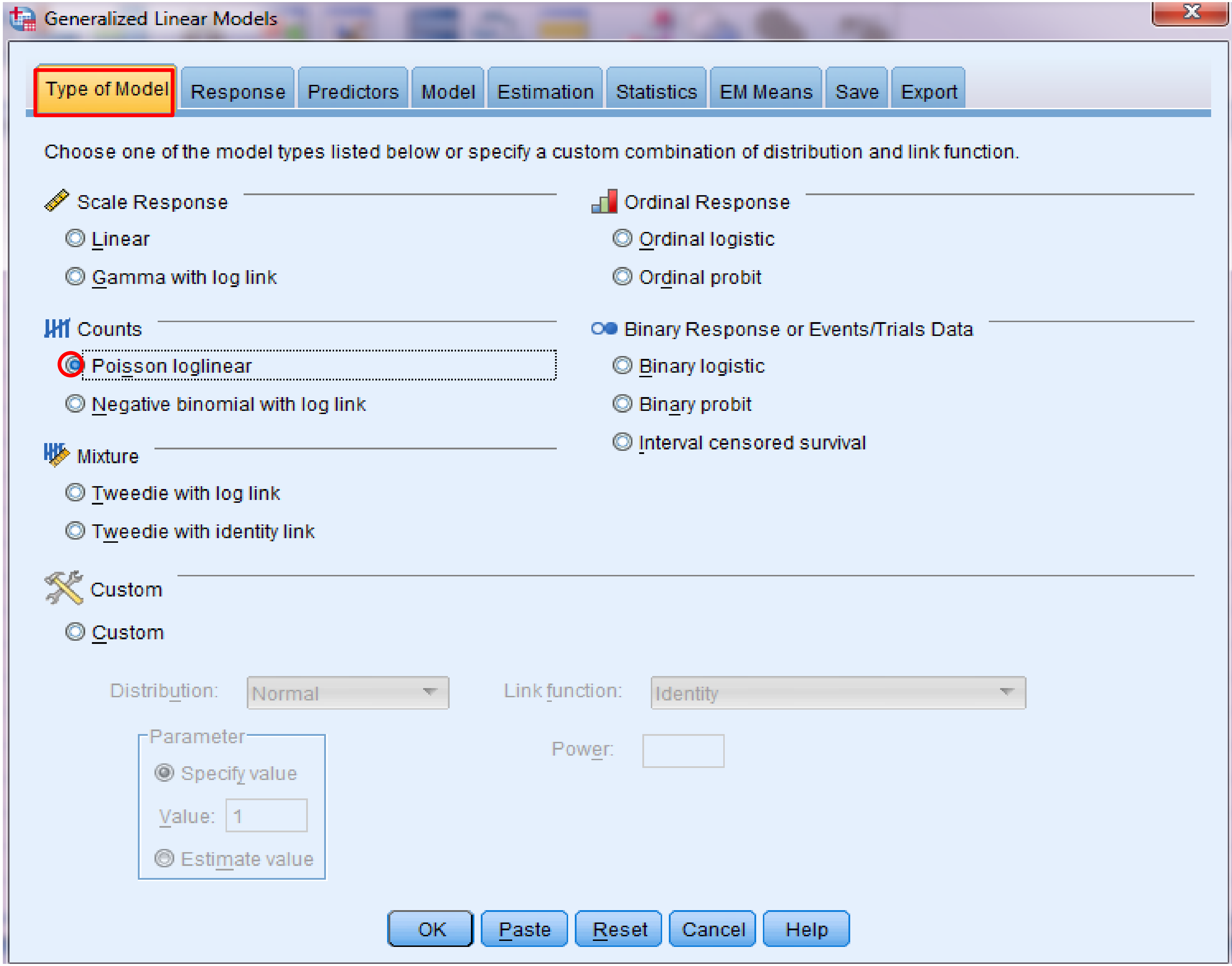
- If you have count as the dependent variable Y,
- Y'= Ln(Y) can be taken to develop a regression model (i.e. Poisson Regression)
- MLE is used to estimate the parameters of regression equation
- If Poisson regression gives Y'= Ln(Y) = a + bx
- Then please understand
  - ✓ If b = 0, then Y'= a i.e. Ln(Y) =a, hence expected / predicted Y = exp(a)
  - ✓ If b ≠ 0, then Y' i.e. Ln(Y) =a + bx , hence expected / predicted Y = exp(a + bx) = exp(a) * exp(bx) and
  - ✓ One unit of x increase, will change Y by exp(b) times (note multiplicative impact not the addition impact that happens in linear regression case)
  - ✓ If b > 0, then Y will become greater by exp(b) times
  - ✓ Like if b=2, then Y will become bigger with each unit increase of x, by exp(2) times
  - ✓ And similarly if b < 0 (like b=-2), then Y will decrease with each unit increase of x, by exp(-2) times

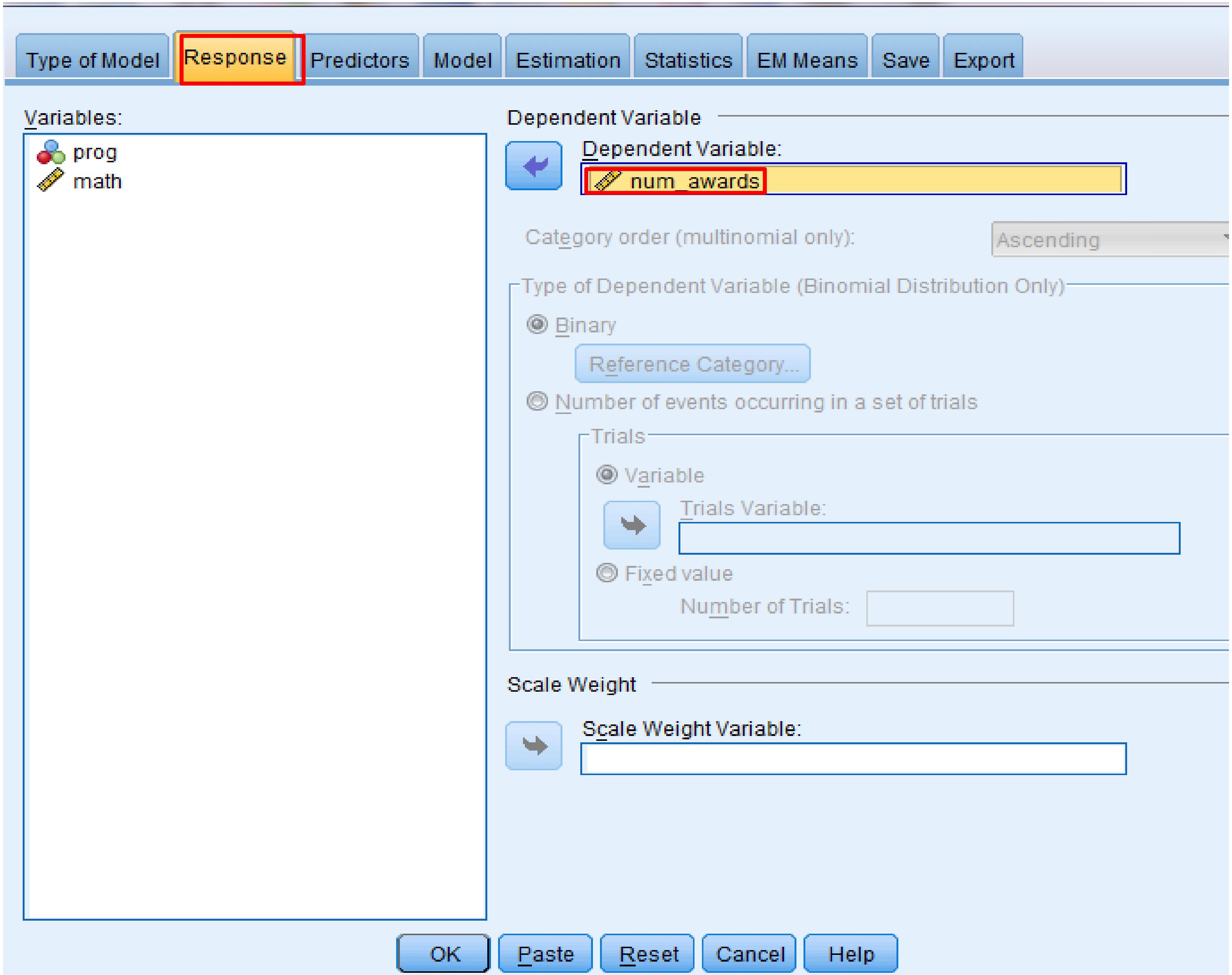
The number of awards earned by students at one high school. Predictors of the number of awards earned include the type of program in which the student was enrolled (e.g., vocational=3, general=1, and academic=2) and the score on their final exam in math.

	num_awards	prog	math
1	0	Vocational	41
2	0	General	41
3	0	Vocational	44
4	0	Vocational	42
5	0	Vocational	40
6	0	General	42
7	0	Vocational	46
8	0	Vocational	40
9	0	Vocational	33
10	0	Vocational	46
11	0	Vocational	40
12	0	Accademic	38
13	0	Vocational	44
14	0	Vocational	37
15	0	Vocational	40
16	0	General	39
17	0	General	43
18	0	Vocational	38
19	0	Accademic	45
20	0	Vocational	39

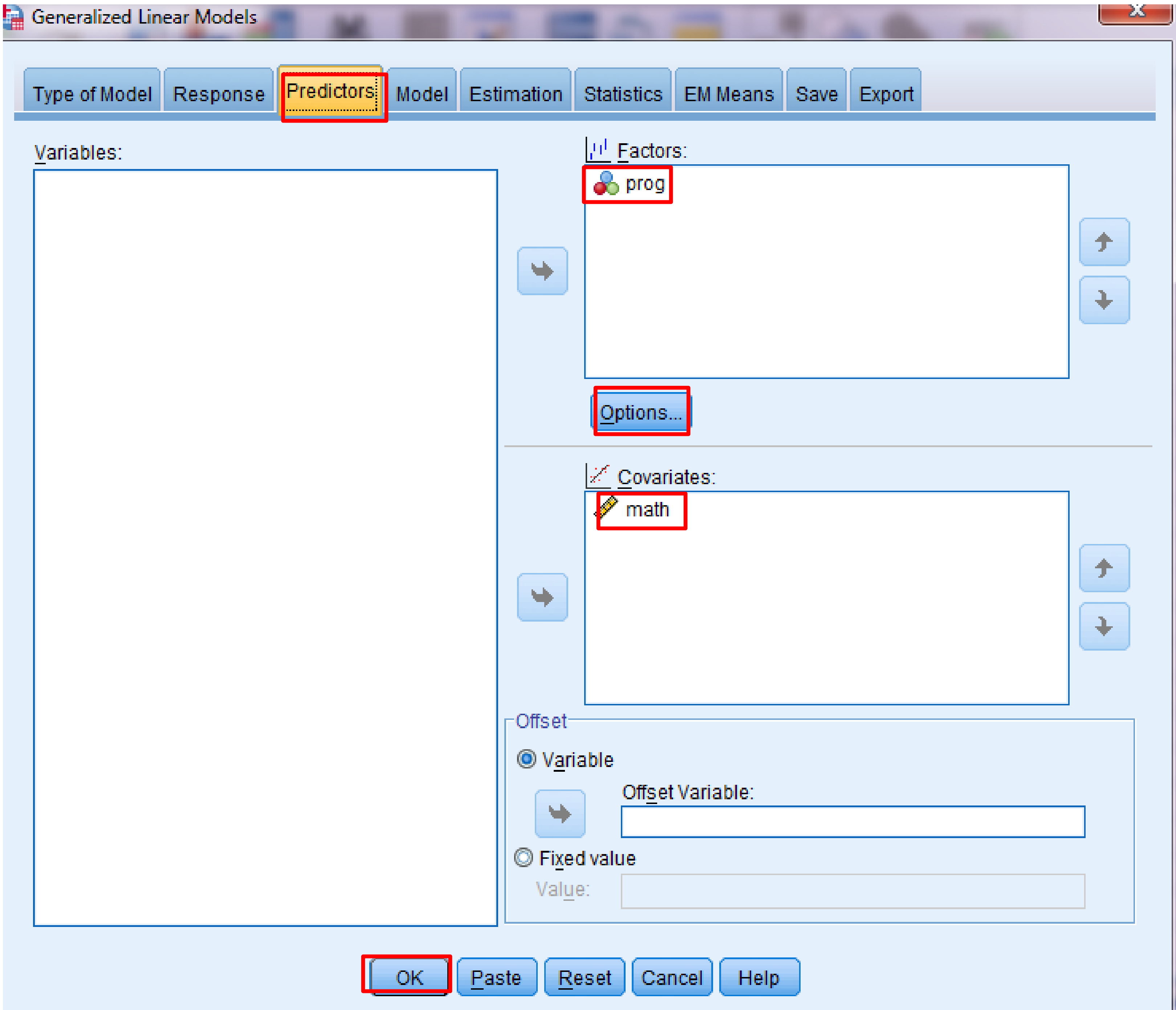


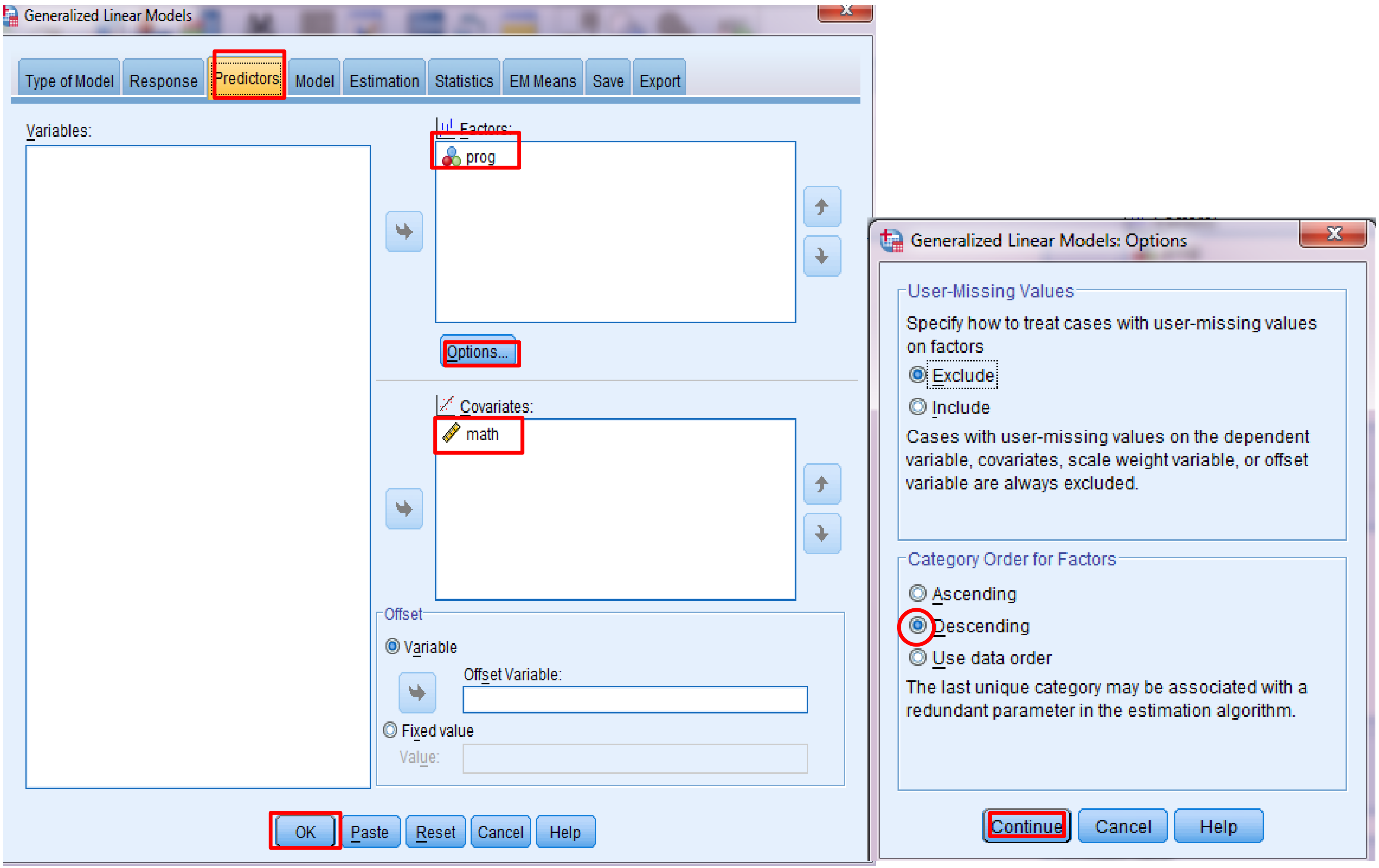
File
Edit
View
Data
Transform
Analyze
Direct Marketing
Graphs
Utilities
Add-ons
Window
Help
num_awards
prog
var
var
var
var
Reports
Descriptive Statistics
Tables
Compare Means
General Linear Model
Generalized Linear Models
Mixed Models
Correlate
Regression
Loglinear
Neural Networks
Classify
Dimension Reduction
Scale
Nonparametric Tests
Forecasting
Survival
Multiple Response
Missing Value Analysis...
Multiple Imputation
Complex Samples
Quality Control
Generalized Linear Models...
Generalized Estimating Equations...


Generalized Linear Models
Type of Model
Response
Predictors
Model
Estimation
Statistics
EM Means
Save
Export
Choose one of the model types listed below or specify a custom combination of distribution and link function.
Scale Response
Linear
Gamma with log link
Ordinal Response
Ordinal logistic
Ordinal probit
Counts
Poisson loglinear
Negative binomial with log link
Binary Response or Events/Trials Data
Binary logistic
Binary probit
Interval censored survival
Mixture
Tweedie with log link
Tweedie with identity link
Custom
Custom
Distribution:
Normal
Link function:
Identity
Parameter
Specify value
Value:
1
Estimate value
Power:
OK
Paste
Reset
Cancel
Help

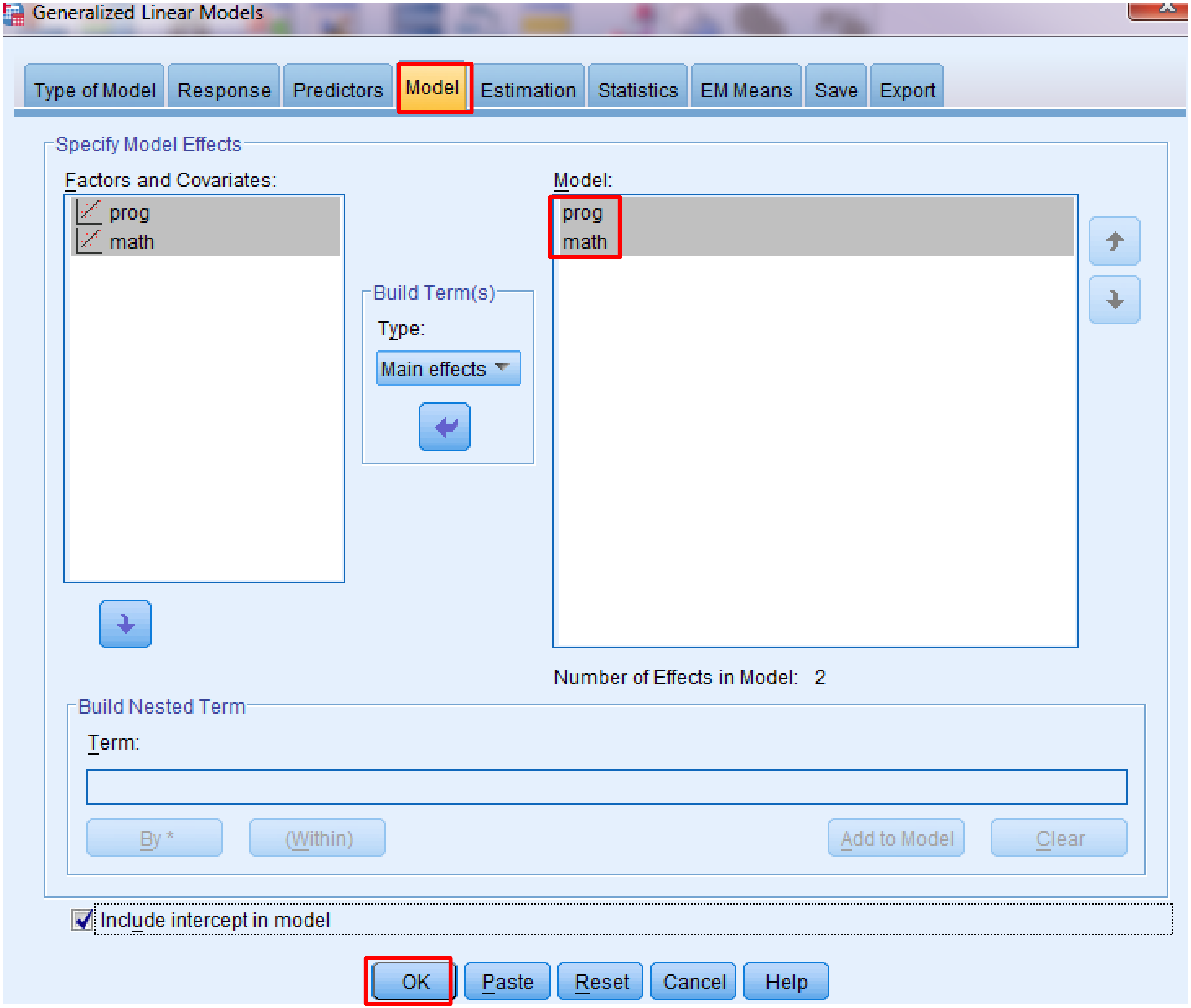


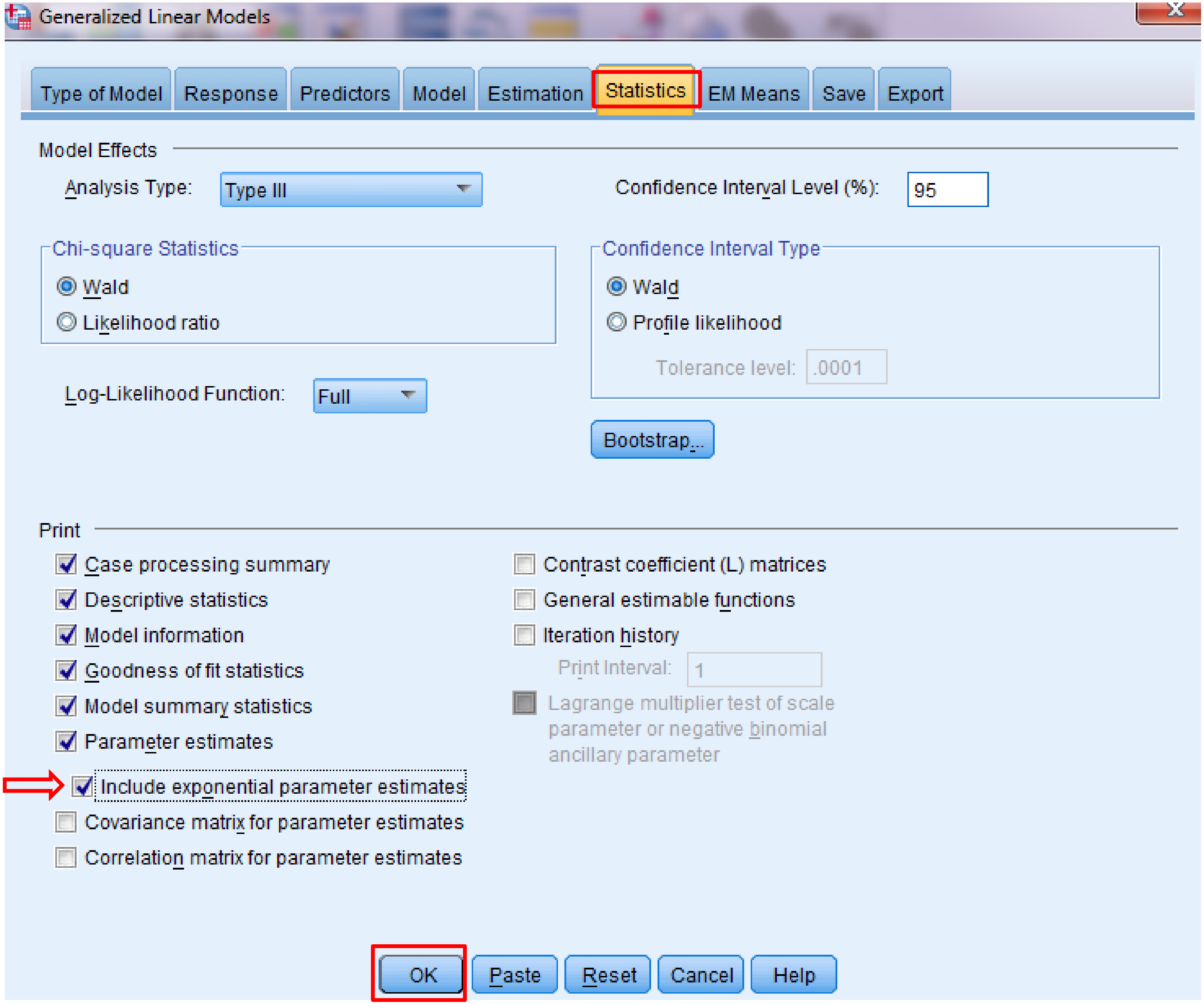
Type of Model
Response
Predictors
Model
Estimation
Statistics
EM Means
Save
Export
Variables:
prog
math
Dependent Variable
Dependent Variable:
num_awards
Category order (multinomial only):
Ascending
Type of Dependent Variable (Binomial Distribution Only)
Binary
Reference Category...
Number of events occurring in a set of trials
Trials
Variable
Trials Variable:
Fixed value
Number of Trials:
Scale Weight
Scale Weight Variable:
OK
Paste
Reset
Cancel
Help


Generalized Linear Models
Type of Model
Response
Predictors
Model
Estimation
Statistics
EM Means
Save
Export
Variables:
Factors:
prog
Options...
Covariates:
math
Offset
Variable
Offset Variable:
Fixed value
Value:
OK
Paste
Reset
Cancel
Help



If you want to **change reference category**, click **Option** and change **Ascending or Descending** icon, then click continue.


Generalized Linear Models
Type of Model
Response
Predictors
Model
Estimation
Statistics
EM Means
Save
Export
Specify Model Effects
Factors and Covariates:
prog
math
Build Term(s)
Type:
Main effects
Model:
prog
math
Number of Effects in Model: 2
Build Nested Term
Term:
By *
(Within)
Add to Model
Clear
Include intercept in model
OK
Paste
Reset
Cancel
Help


Generalized Linear Models
Type of Model
Response
Predictors
Model
Estimation
Statistics
EM Means
Save
Export
Model Effects
Analysis Type:
Type III
Confidence Interval Level (%):
95
Chi-square Statistics
Wald
Likelihood ratio
Confidence Interval Type
Wald
Profile likelihood
Tolerance level:
.0001
Log-Likelihood Function:
Full
Bootstrap...
Print
Case processing summary
Descriptive statistics
Model information
Goodness of fit statistics
Model summary statistics
Parameter estimates
Include exponential parameter estimates
Covariance matrix for parameter estimates
Correlation matrix for parameter estimates
Contrast coefficient (L) matrices
General estimable functions
Iteration history
Print Interval:
1
Lagrange multiplier test of scale parameter or negative binomial ancillary parameter
OK
Paste
Reset
Cancel
Help

Omnibus Test[a]

Likelihood Ratio Chi-Square	df	Sig.
58.234	2	.000

Dependent Variable: num_awards
Model: (Intercept), prog, math

a. Compares the fitted model against the intercept-only model.

The likelihood ratio chi-square test indicates that the full model was a significant improvement in fit over a null (no predictors) model ($p<.001$).

Parameter Estimates

Parameter	B	Std. Error	95% Wald Confidence Interval		Hypothesis Test			Exp(B)	95% Wald Confidence Interval for Exp(B)	
			Lower	Upper	Wald Chi-Square	df	Sig.		Lower	Upper
(Intercept)	-5.247	.6585	-6.538	-3.957	63.503	1	.000	.005	.001	.019
[prog=3]	.370	.4411	-.495	1.234	.703	1	.402	1.447	.610	3.436
[prog=2]	1.084	.3583	.382	1.786	9.153	1	.002	2.956	1.465	5.966
[prog=1]	0[a]	.	.	.	.	.	.	1	.	.
math	.070	.0106	.049	.091	43.806	1	.000	1.073	1.051	1.095
(Scale)	1[b]									

Dependent Variable: num_awards
Model: (Intercept), prog, math

a. Set to zero because this parameter is redundant.

b. Fixed at the displayed value.

Score of math is a significant predictor of the number of awards earned by students (B=0.70, p<0.001). For one unit increase in math score, the predicted log counts of number of awards earned by students increased 0.07 times. The incidence rate ratio (IRR) indicates that for every one unit increase on the predictor (math score), the incidence rate (for number of award) increased by a factor of 1.073 (exp(0.07)) or (1.073-1)*100= 7.3%.

Parameter Estimates

Parameter	B	Std. Error	95% Wald Confidence Interval		Hypothesis Test			Exp(B)	95% Wald Confidence Interval for Exp(B)	
			Lower	Upper	Wald Chi-Square	df	Sig.		Lower	Upper
(Intercept)	-5.247	.6585	-6.538	-3.957	63.503	1	.000	.005	.001	.019
[prog=3]	.370	.4411	-.495	1.234	.703	1	.402	1.447	.610	3.436
[prog=2]	1.084	.3583	.382	1.786	9.153	1	.002	2.956	1.465	5.966
[prog=1]	0[a]	.	.	.	.	.	.	1	.	.
math	.070	.0106	.049	.091	43.806	1	.000	1.073	1.051	1.095
(Scale)	1[b]									

Dependent Variable: num_awards
Model: (Intercept), prog, math

a. Set to zero because this parameter is redundant.

b. Fixed at the displayed value.

Program (Academic Vs General) is a significant predictor of the number of awards earned by students (B=1.08, p<0.01). The predicted log counts of number of awards earned for academic program, the incidence rate (the number of awards earned) was 2.956 times greater than that of general program. In other words, the IR for academic program was (2.95-1)*100=195% greater than that of general program.